

PRSQ-02-IMDB Movies

1. Abstract

The Predictive Recommendation System using SQL Queries (PRSQ) is an advanced analytical framework designed to extract actionable insights from structured datasets. Leveraging SQL-based querying techniques, this system facilitates intelligent data exploration, trend identification, and predictive modelling. The project focuses on transforming raw data into meaningful information that supports strategic decision-making in modern data-driven environments.

2. Introduction

In the era of big data, organizations require efficient systems to process and interpret vast volumes of information. PRSQ aims to bridge the gap between raw data storage and intelligent insights by utilizing structured query language (SQL) to perform dynamic data analysis.

This project demonstrates how relational databases can be used not only for storage but also for predictive and recommendation-based analytics.

3. Objectives

- To design an efficient SQL-based data analysis system
- To perform complex queries for extracting meaningful insights
- To enable predictive recommendations using structured data
- To improve decision-making through data visualization and interpretation
- To demonstrate scalable and future-ready database solutions

4. Methodology

The PRSQ system follows a structured workflow:

4.1 Data Collection

Data is collected from structured datasets (e.g., IMDB movie dataset).

4.2 Data Cleaning

Ensuring consistency, removing null values, and formatting data.

4.3 Database Design

- Creation of relational tables
- Establishing primary and foreign key relationships

4.4 Query Implementation

- Use of SELECT, JOIN, GROUP BY, HAVING
- Advanced SQL queries for insights

4.5 Analysis & Prediction

- Trend analysis
 - Recommendation logic based on query outputs
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5. Tools & Technologies

- SQL (MySQL / PostgreSQL / SQL Server)
 - Database Management Systems
 - Data Visualization Tools (optional)
 - Microsoft PowerPoint / Word
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6. Key Features

- Scalable and efficient query execution
 - Real-time data analysis capability
 - Structured and modular design
 - Future-ready integration with AI/ML systems
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7. Results & Discussion

The system successfully demonstrates how SQL queries can be used to:

- Identify trends in datasets
- Extract high-value insights
- Generate recommendations

This approach highlights the potential of SQL beyond basic data retrieval.

8. Future Scope

- Integration with Machine Learning models
 - Real-time recommendation engines
 - Cloud-based database deployment
 - AI-powered predictive analytics
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9. Conclusion

The PRSQ system represents a forward-thinking approach to data analysis using SQL. It emphasizes efficiency, scalability, and adaptability, making it suitable for modern enterprise applications.

Perform the Problem Queries:

- a) Can you get all data about movies?
`SELECT * FROM movies;`
- b) How do you get all data about directors?
`SELECT * FROM directors;`
- c) Check how many movies are present in IMDB.
`SELECT COUNT(*) AS total_movies FROM movies;`
- d) Find these 3 directors: James Cameron ; Luc Besson ; John Woo
`SELECT * FROM directors
WHERE name IN ('James Cameron', 'Luc Besson', 'John Woo');`
- e) Find all directors with name starting with S.
`SELECT * FROM directors WHERE name LIKE 'S%';`
- f) Count female directors.
`SELECT COUNT(*) AS female_directors
FROM directors
WHERE gender = 1;`
- g) Find the name of the 10th first women directors?
`SELECT name FROM directors WHERE gender = 1 ORDER BY name ASC LIMIT 1
OFFSET 9;`
- h) What are the 3 most popular movies?
`SELECT m.title, m.popularity, d.name AS director_name FROM movies m JOIN
directors d ON m.director_id = d.id ORDER BY m.popularity DESC LIMIT 3;`
- i) What are the 3 most bankable movies?
`SELECT m.title, m.revenue, d.name AS director_name FROM movies m JOIN
directors d ON m.director_id = d.id ORDER BY m.revenue DESC LIMIT 3;`
- j) What is the most awarded average vote since the January 1st, 2000?

```
SELECT m.title, m.vote_average, m.release_date, d.name AS director_name  
FROM movies m JOIN directors d ON m.director_id = d.id WHERE  
m.release_date >= '2000-01-01' ORDER BY m.vote_average DESC LIMIT 1;
```

k) Which movie(s) were directed by Brenda Chapman?

```
SELECT m.title, m.release_date, m.vote_average FROM movies m JOIN  
directors d ON m.director_id = d.id WHERE d.name = 'Brenda Chapman';
```

l) Which director made the most movies?

```
SELECT d.name, COUNT(m.id) AS movie_count FROM directors d JOIN movies  
m ON d.id = m.director_id GROUP BY d.id, d.name ORDER BY movie_count  
DESC LIMIT 1;
```

m) Which director is the most bankable?

```
SELECT d.name, SUM(m.revenue) AS total_revenue FROM directors d JOIN  
movies m ON d.id = m.director_id GROUP BY d.id, d.name ORDER BY  
total_revenue DESC LIMIT 1;
```